FuseLID: Vision-Language Model-Based Camera-Radar Fusion for Liquid Identification

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ABSTRACT

Service robots frequently perform liquid-related tasks such as pouring, which necessitate accurate liquid identification. Vision-language models (VLM) demonstrate excellent capabilities in object recognition but face limitations in distinguishing visually similar liquids. As a promising solution, millimeter-wave (mmWave) radar combined with neural networks has been widely adopted for material recognition and liquid identification. However, training a radar-based liquid classifier requires extensive labeled radar-liquid data pairs and struggles with generalization to real-world environments. To address this challenge, we propose FuseLID, a VLM-based camera-radar late-fusion system designed to enhance liquid identification. FuseLID leverages the weak penetration property of mmWave signals and integrates them with the precise motion control of a robot to acquire radar representations of different liquids. Subsequently, a VLM-based late-fusion module combines camera and radar outputs to achieve enhanced liquid identification with limited radar-liquid data. Preliminary experiments show that FuseLID improves liquid identification accuracy from 51.6% to 95.7% when classifying several commonly consumed beverages, including visually similar ones.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**.

KEYWORDS

Liquid identification, vision language model, mmWave radar, camera-radar fusion.

1 INTRODUCTION

Service robots, an important category of robotics, frequently handle liquid-related tasks like pouring or refilling a cup of drink, requiring accurate liquid identification. Vision-language models (VLM) exhibit excellent capabilities to recognize objects across diverse workspaces, but they exhibit fundamental limitations in distinguishing visually similar liquids (e.g., color, transparency). Our simple test in Figure 1 and prior studies consistently demonstrate VLM failure due to the lack of distinctive optical features in such scenarios [3, 5, 6].

To overcome this optical perception gap, researchers investigate millimeter-wave (mmWave) radar sensing. Radar signals are sensitive to a material's dielectric properties, which vary significantly across common materials [2]. Distinct from lower frequency signals, mmWave signals possess shorter wavelengths that lead to minimal penetration and stronger surface scattering within liquids, amplifying differences in their radar reflection signatures [8]. Consequently, mmWave radar-based liquid identification has advanced substantially, yielding promising results [8, 10, 13]. However, existing methods depend on extensive radar-liquid data pairs, which hinders their real-world deployment. In this context, we ask: Can

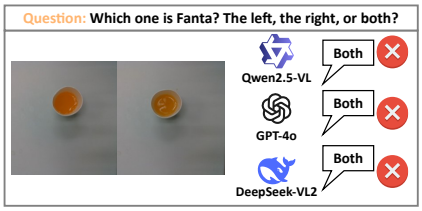


Figure 1: VLMs frequently misclassify visually similar liquids such as Fanta (left) and orange juice (right), highlighting their limitations in fine-grained liquid recognition.

we leverage the robust semantic understanding capabilities of VLM and integrate them with limited material-sensitive mmWave radar to achieve fine-grained liquid identification?

While conceptually feasible, developing a VLM and radar fusion system faces several challenges: (1) Heterogeneous Fusion: The fundamentally distinct characteristics of VLM-extracted image features and mmWave radar data necessitate robust fusion strategies. Additionally, VLM cannot directly process raw radar data. (2) Scarce Paired Datasets: Currently, no publicly available, large-scale datasets exist that pair RGB images with mmWave radar data for liquid identification. Furthermore, sparse radar reflection points fail to fully capture liquid characteristics.

To overcome these hurdles, we present FuseLID, the robotic system designed to late-fusion VLM results and radar data for enhanced liquid identification, as illustrated in Figure 2. FuseLID exploits the robot's precise motion to synthesize a large antenna aperture, significantly enhancing radar signal resolution and information richness, helping overcome data sparsity. In response to heterogeneous fusion and scarce paired data, FuseLID employs a late-fusion architecture. It first trains a radar classification network independently using only our self-built radar-liquid dataset, eliminating the need for paired visual inputs during radar model training. Subsequently, the framework leverages the VLM's reasoning capabilities to analyze candidate results from visual and radar-based classifiers, improving liquid classification accuracy by exploiting the complementary strengths of cross-modal data. This design enables good identification results to be achieved even with only a limited Radar-Liquid dataset training. Experimental validation on a real-world platform demonstrates that FuseLID achieves significant performance improvements over VLM-only and radar-only baselines in distinguishing visually or chemically similar liquids. This validates its practical efficacy in liquid identification.

2 RELATED WORK

Accurate liquid identification, especially within domestic settings, plays a vital role in facilitating context-aware proactive interactions for service robots based on user requirements and activities. Leveraging the massive datasets of VLM, vision-based liquid detection methods have developed diverse theoretical frameworks and

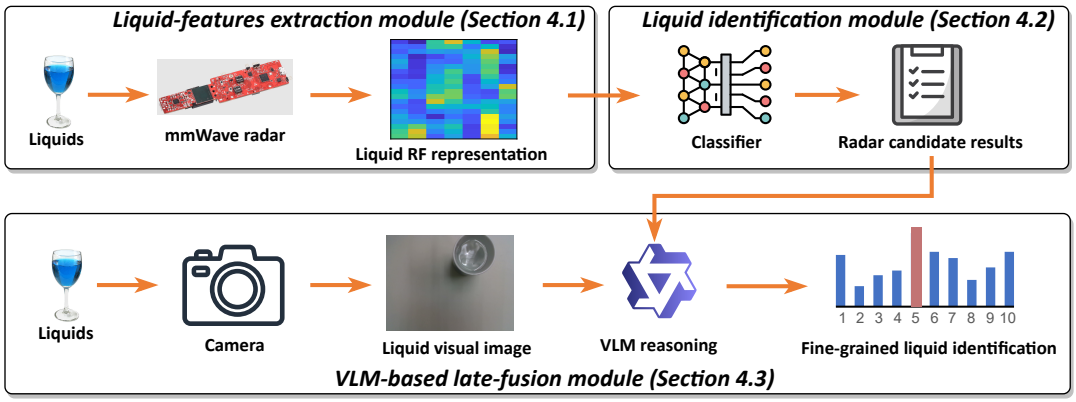


Figure 2: The architecture of FuseLID.

meaningful applications. However, these methods impose strict requirements on lighting conditions, liquid colors, and transparency levels, often failing to distinguish similar liquids. Complementing visual techniques, radio frequency (RF) sensor-based methods have emerged targeting robust liquid recognition by characterizing material-specific dielectric properties and other intrinsic parameters [4, 7, 9, 11–13]. Specifically, Tagscan utilizes received signal strength indicators (RSSI) and phase variations obtained through radio frequency identification (RFID) interrogations to build classification models for 10 distinct liquid categories [12]. FG-LiquID designs a novel neural network for liquid sensing using millimeter-wave radar, enabling fine-grained identification of 30 different liquid types [7]. Nevertheless, creating comprehensive repositories encompassing extensive solution varieties proves practically challenging.

Distinct from conventional approaches, our proposed FuseLID implements a late-fusion architecture that synergistically combines vision-language models with radar sensing for enhanced liquid recognition accuracy. The FuseLID framework is designed to use VLM-derived visual cues to refine predictions from dedicated radar classifiers through semantic guidance. By exploiting the pretrained knowledge embedded in VLM, our approach demonstrates robust performance even with limited radar samples, particularly excelling in distinguishing visually similar liquids.

3 MOTIVATION OF LATE-FUSION

Multi-modal fusion strategies are typically classified as early-fusion, middle-fusion, or late-fusion, all of which can effectively extract features from different modalities. We adopt late-fusion primarily due to 1) data dimension discrepancies. Typical visual data (640×480) and our radar representation (7×20) exhibit significant dimensional mismatches. This scale disparity makes network design for comprehensive feature extraction in early/middle-fusion architectures technically impractical. 2) Dataset size limitations. Although directly training a radar model for liquid identification is a straightforward solution, it requires more than ten thousand Radar-Liquid data pairs [15]. Late-fusion can fully utilize prior knowledge from VLM to further optimize and filter radar model outputs, achieving performance comparable to large-scale dataset training with only limited data usage.

4 SYSTEM DESIGN

FuseLID consists of three major modules, as illustrated in Figure 2. First, a liquid-features extraction module employs a beamforming algorithm to obtain the RF representation of liquids. Second, a liquid identification module leverages a self-built Radar-Liquid dataset to train a neural network. Finally, the VLM-based late-fusion module integrates two modalities during the network inference phase by harnessing pre-trained VLM datasets to select reliable results from the radar classifier network, thereby improving the accuracy of liquid identification.

4.1 Liquid-Features Extraction Module

As noted in [14], the orientation of the radar antenna significantly affects the quality of material feature extraction. To address this, we mounted the mmWave radar on the manipulators of the service robot to position it above the containers and orient it toward the liquid automatically. Since mmWave signals result in minimal penetration and strong surface scattering within liquids, the dominant path of the received radar signal encapsulates the liquid's RF features. Consequently, in this module, we employ a beamforming algorithm to identify the dominant path and extract the corresponding reflected signal as the RF representation of the liquid.

4.1.1 Beamforming Algorithm. Let θ_l and τ_l denote the angle of arrival (AoA) and time of flight (ToF) of the lth path, respectively. We assume the number of receiver antennas is M, arranged in a uniform linear array, and the radio signal is transmitted over K subcarriers. The phase shift vector can be expressed as

$$a(\theta_l, \tau_l) = [1, \cdots, \phi_{mk}(\theta_l, \tau_l), \cdots, \phi_{MK}(\theta_l, \tau_l)]^T, \quad (1)$$

where $\phi_{mk}(\theta_l, \tau_l)$ represents the phase shift at the kth subcarrier sample and the mth antenna of lth path [1, 16].

Assuming L paths exist in the environment, the steering matrix is defined as $A = [a(\theta_1, \tau_1), a(\theta_2, \tau_2), \cdots, a(\theta_L, \tau_L)]$. The received signal matrix is given by $H(t) = As(t) + n(t)$, where $s(t)$ is the emitted mmWave signal and $n(t)$ denotes background noise.

To extract the reflected paths, we discretize the AoA-ToF space into a grid, defining the corresponding steering matrix as

$$\hat{A} = [a(\hat{\theta}_1, \hat{\tau}_1), a(\hat{\theta}_2, \hat{\tau}_2), \cdots, a(\hat{\theta}_N, \hat{\tau}_N)], \quad (2)$$

where N represents the number of AOA-TOF grid points.

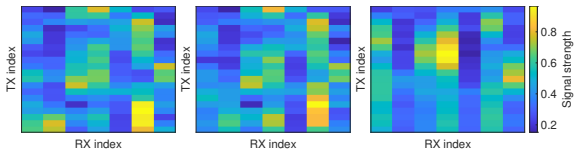


Figure 3: RF representation of three liquids. Fanta (left) and Sprite (middle) appear similar, whereas water (right) is clearly distinct.

Then, the AoA-ToF spectrum is computed by projecting $\hat{A}$ onto the received data $H(t)$, i.e.,

$$P = \hat{A}^H H(t) = \hat{A}^H A s(t) + \hat{n}(t), \tag{3}$$

where $A \in C^{MK \times L}$, $\hat{A} \in C^{MK \times N^2}$, $s \in C^{L \times 1}$, and $\hat{n}(t) = \hat{A}^H n(t)$.

According to Eq. (3), with a dense AoA-ToF grid, i.e., $N \gg L$, the AoA-ToF spectrum P will have large amplitude on the grid points corresponding to the reflected paths. The AoA-ToF of the dominant path is the one with the maximum amplitude by excluding the line-of-sight (LOS) path, i.e.,

$$(\theta^*, \tau^*) = \arg \max_{(\hat{\theta}_i, \hat{\tau}_i) \neq (\theta_0, \tau_0)} |P| = |\hat{A}^H H(t)|, \tag{4}$$

where (θ_0, τ_0) denotes the AoA-ToF of the LOS path. Then, the signal from the dominant path can be expressed as

$$\tilde{s}_d(t) = a(\theta^*, \tau^*)^H H(t). \tag{5}$$

4.1.2 RF Representation of Liquids. Recall the Friis transmission formula is given by

$$A_d = A_0 g_t g_r \frac{\lambda}{4\pi R} \Gamma_{liquid}, \tag{6}$$

where A_0 and A_d denote the amplitude of the transmitted and received signal at propagation distance R, respectively. Here, g_t and g_r represent the transmit and receive antenna gains, λ is the wavelength, and Γ_{liquid} is the reflection coefficient of the liquid.

With all other parameters fixed, the signal amplitude is directly proportional to the liquid's reflection coefficient. Therefore, the RF representation for any liquid can be derived using $\tilde{s}_d(t)$ and (θ^*, τ^*) obtained by Eq. (4) and (5). Meanwhile, with the robot's precise motion to synthesize a large antenna aperture, we can obtain a 7 × 20 Tx-Rx pairs channel matrix to represent distinguish liquid precisely. Figure 3 illustrates several liquids' RF representation.

4.2 Liquid Identification Module

By leveraging the liquid-feature extraction module, we obtain the RF representation corresponding to different liquid types. In this subsection, we propose a U-Net architecture liquid identification module comprising convolutional neural networks (CNNs) and skip connection to classify liquids from radar signals.

4.2.1 Network Architecture. The radar encoder employs a hierarchical CNN architecture to extract high-level latent representations from 2D radar inputs. The encoder is composed of stacked basic blocks, each comprising a 2D convolutional layer, batch normalization, and a ReLU activation function. The number of channels increases progressively in deeper layers to capture richer and more diverse features. The decoder adopts a symmetrical design to the encoder, with a decreasing number of channels and increasing output dimensions across stacked blocks. Dropout is applied to mitigate overfitting during high-dimensional expansion. A linear layer outputs logits for four classes using softmax activation.

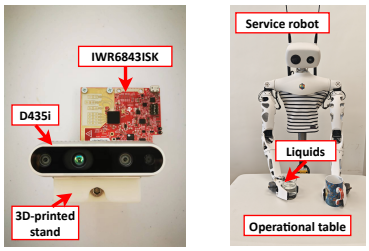


Figure 4: Hardware implementation (left) and the service robot workspace (right).

4.2.2 Loss Function Design. To train the network for liquid identification from radar feature matrices, we adopt cross-entropy as the loss function. Cross-entropy directly quantifies the discrepancy between the model's output probability distribution and the ground truth labels. In multi-class classification tasks, it effectively guides model optimization and enhances training efficiency.

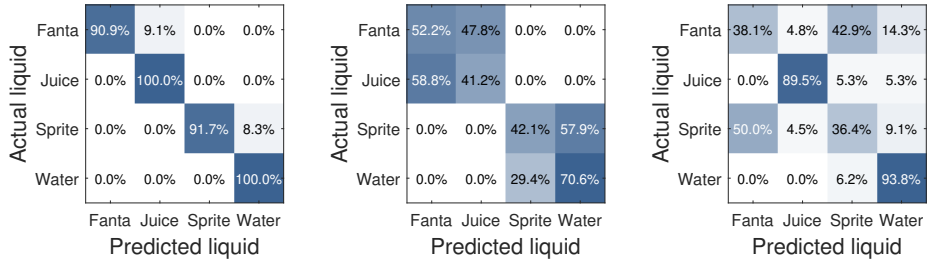
4.2.3 Self-built Radar-Liquid Dataset. We made a new dataset comprising RF representations of liquids. The dataset consists of 400 realistic radar-liquid data pairings, encompassing four commonly encountered liquids in daily life: Fanta, orange juice, Sprite, and water. Each sample is carefully curated to include one liquid category with variations in cup types and liquid levels, ensuring representation of diverse environments where service robots are prevalent.

4.3 VLM-Based Late-Fusion Module

As discussed in Section 1, while designing a comprehensive end-to-end network is a straightforward approach for liquid identification, it requires extensive labeled data for training. To address this limitation, this subsection proposes a visual-radar late-fusion framework to enable robust liquid identification with limited radar data.

We exploit the pre-trained VLM's ability to extract semantic information from visual data, which provides prior knowledge about liquid properties. Although VLM struggle to distinguish visually similar liquids, their capacity to differentiate liquids with distinct visual features allows for refining the liquid identification module's predictions, significantly reducing the data requirements for training. The output format of the liquid identification module was modified to generate both the most probable candidate predictions and their associated confidence scores. Subsequently, the late-fusion approach integrates candidate results from the radar classifier and visual data into the VLM, leveraging the VLM's reasoning capabilities to analyze outcomes and improve liquid classification accuracy.

To validate this approach, we conducted a preliminary experiment involving a Fanta beverage on a table. The classifier was trained on 200 samples (50 per category: Fanta, orange juice, Sprite, and water), followed by direct prediction using extracted radar features. Due to the limited dataset size, the network experienced severe overfitting during training, resulting in poor generalization during testing. Consequently, the network classified the orange-colored liquid as either Fanta or Sprite, a result consistent with



(a) FuseLID (Our). (b) VLM-only. (c) Radar-only.

Figure 5: Confusion matrices of liquid identification.

expectations, as these weakly acidic liquids exhibit similar radar features. Meanwhile, the VLM identified the orange-colored liquid as either Fanta or orange juice via visual analysis, assigning higher confidence to Fanta. Through the late-fusion and VLM reasoning framework, the system ultimately classified the liquid as Fanta.

5 EXPERIMENTAL EVALUATION

5.1 Experiment Setup

In this section, we evaluate FuseLID using different liquids. We implement FuseLID using a TI IWR6843ISK mmWave radar, which operating at a center frequency of 61.8 GHz and a bandwidth of 3.6 GHz. The mmWave radar unit is mounted alongside an Intel Realsense D435i RGB-D camera on a custom-designed 3D-printed stand, fixed on the hand of Reachy robot, as shown in Figure 4. With robot's precise motion to synthesize a large antenna aperture, we can obtain a 7×20 Tx-Rx pairs. The entire system operates on Ubuntu 20.04 and interfaces with the devices through Ethernet connections for robust data transmission and control.

5.2 Overall Performance

We now present the liquid identification performance of FuseLID, and further compare it with a VLM-only [1] and a radar-only identification algorithm. We prepared four liquids that appear in our self-built dataset: Fanta, orange juice, Sprite, and water. We use confusion matrices in Figure 5 to show the identification performance of different methods. We observe that the late-fusion solution used in FuseLID is the most robust compared with the VLM-only and radar-only methods. The overall accuracy for identifying the liquids is 95.7%, substantially higher than the VLM-only (51.6%) and radar-only (64.5%) methods. Notably, FuseLID prioritizes visual data during the VLM-based late-fusion phase, regarding radar candidates as supplementary information. Despite these advancements, visually similar liquids remain susceptible to misclassifications.

Specifically, VLM-only methods enable coarse-grained correct classification of liquids based on color and transparency, yet fail to achieve fine-grained liquid identification, particularly for visually similar liquids. Figure 5(b) presents the confusion matrix of the VLM-only method. Benefiting from the extensive pretraining data of VLM, it can be observed that the VLM-only method correctly identifies transparent liquids as Sprite or water, yet cannot distinguish between specific liquid types. Figure 5(c) demonstrates the confusion matrix of the radar-only approach. Here we employed 400 training samples, showing that radar-only methods easily distinguish juice from water, yet cannot differentiate Fanta and Sprite which share similar acidic properties.

[1]We use Qwen2.5-VL-7B model and visual data for liquid identification.

6 LIMITATIONS AND FUTURE WORK

Category Limitations. The current work focuses on liquid recognition for only 4 categories, fewer than prior studies, and limited radar-liquid data pairs also constrain FuseLID's performance. In future work, we plan to collect more liquid categories and extend liquid identification to over 16 types, further enhancing practical applicability.

Optimizing the Late-Fusion Strategy. As demonstrated in Section 5.2 and Figure 5(a), the current fusion strategy prioritizes visual data, which can lead to identification errors when visual and radar-based identification results conflict. In future work, we aim to refine the late-fusion strategy by leveraging the VLM's reasoning capabilities through techniques such as low-rank adaptation (LoRA) fine-tuning. This will enable FuseLID to more effectively reason visual and radar outcomes to further improve the performance.

7 CONCLUSION

Reliable liquid identification is essential for enabling service robots to perform complex tasks effectively. While VLM demonstrates strong capabilities in object detection, their performance declines when distinguishing visually similar liquids. Although mmWave radar has shown potential for liquid sensing, its practical deployment is hindered by data-intensive training. Our FuseLID framework addresses these limitations through an innovative design. The framework enables robust extraction of RF liquid representations from radar data. A key contribution is the use of VLM's reasoning to optimize cross-modal integration, achieving fine-grained liquid identification. Preliminary experimental results demonstrate that FuseLID achieves a substantial accuracy improvement (from 51.6% to 95.7%) in identifying visually similar liquids, surpassing existing methodological constraints and offering a practical, efficient solution for precise liquid perception in challenging, data-scarce scenarios.

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